

Work must be shown to receive credit. All problems are of equal value.

1. Let  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}$  be a set of  $n + 1$  vectors in  $\mathbb{R}^n$ . The which of the following is true? Explain.

- (A)  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}$  is a basis for  $\mathbb{R}^n$ .
- (B)  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}$  is linearly independent.
- (C)  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}$  is linearly dependent .
- (D) The  $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1})$  is not a subspace of  $\mathbb{R}^n$ .

2. If the reduced row echelon form of a system is  $\left[ \begin{array}{ccc|c} 1 & 0 & 3 & 29 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right]$ . Then the system has,

- (A) a unique solution.
- (B) no solution.
- (C) infinitely many solutions.
- (D) two solutions.

3. If the *reduced row echelon form* of the augmented matrix of a system of linear equations has a leading coefficient 1 in the last column then the system has:

- (A) a unique solution.
- (B) no solution.
- (C) infinitely many solutions.
- (D) none of the above.

4. If  $\det A = 6$  and  $AB = \begin{bmatrix} -2 & 0 & 0 \\ 3 & -1 & 0 \\ 1 & 5 & 1 \end{bmatrix}$  find  $\det B$ .

5. Let  $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 3 & 2 \\ 0 & 2 & 1 \end{bmatrix}$ . Find the characteristic polynomial of  $A$ .

6. Let  $A'$  be the reduced row echelon form of a matrix  $A$ . Then which of the following is true.

- (A)  $\det A = \det A'$
- (B)  $\det A = \pm \det A'$
- (C)  $\det A = \lambda \det A'$  for some nonzero scalar  $\lambda$ .
- (D)  $\det A$  and  $\det A'$  are unrelated.

7. Let  $A$ ,  $B$ , and  $C$  be three matrices such that  $AB = CA$  and  $A$  is invertible. Then,

- (A)  $\det B = \det C$
- (B)  $\det B = \pm \det C$
- (C)  $\det B = -\det C$
- (D)  $\det B$  and  $\det C$  are unrelated.

8. Let  $A$  be an  $m \times n$  real matrix. Then the function  $T[\mathbf{x}] = A\mathbf{x}$  is

- (A) a linear map from  $\mathbb{R}^m$  to  $\mathbb{R}^n$ .
- (B) a non-linear map from  $\mathbb{R}^m$  to  $\mathbb{R}^n$ .
- (C) a linear map from  $\mathbb{R}^n$  to  $\mathbb{R}^m$ .
- (D) not a function.

9. Let  $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ . Then which of the following is true.

- (A)  $A$  is singular.
- (B)  $A$  is invertible and  $A^{-1} = \begin{bmatrix} 1 & 1/4 & 1/7 \\ 1/2 & 1/5 & 1/8 \\ 1/3 & 1/6 & 1/9 \end{bmatrix}$ .
- (C)  $\det A = 1$ .
- (D)  $\det A = 9$ .

10. Let  $\mathbf{v} = (3, -2, 4)$ ,  $\mathbf{w} = (-3, 2, -4)$ , and  $\mathbf{u} = (-6, 4, -8)$  be three vectors in  $\mathbb{R}^n$ . Find the dimension of  $\text{Span}(\mathbf{v}, \mathbf{w}, \mathbf{u})$  and explain your answer.

11. Let  $A$  be an  $n \times n$  real matrix. If the equation  $A\mathbf{x} = \mathbf{0}$  has a trivial solution then,

- (A)  $A$  is invertible.
- (B)  $A$  is singular.
- (C) the system has infinitely many solutions.
- (D) none of the above.

12. Let  $A$  be a matrix with  $\det A = 3$ .

- (A)  $\det(A^{-1}) = 1/3$
- (B)  $\det(A^{-1}) = 1/3^2$
- (C)  $\det(A^{-1})$  is undefined because  $A^{-1}$  might not exist.
- (D)  $\det(A^{-1})$  and  $\det A$  are unrelated.

13. Let  $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 10 & 2 & 0 & 0 \\ -7 & -1 & 3 & 0 \\ 25 & 5 & 11 & 5 \end{bmatrix}$  find  $\det A$ .

14. Let  $A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$ . Find  $A^{-1}$  if it exists.

15. Find the inverse of the matrix  $A = \begin{bmatrix} 4 & -3 & 0 & 0 \\ -5 & 4 & 0 & 0 \\ 0 & 0 & 3 & -2 \\ 0 & 0 & 5 & -3 \end{bmatrix}$  if it exists.